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Performance Analysis of Universal Filtered Multicarrier waveform with various design parameters for 5G and beyond wireless Networks

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Evolution of cellular networks

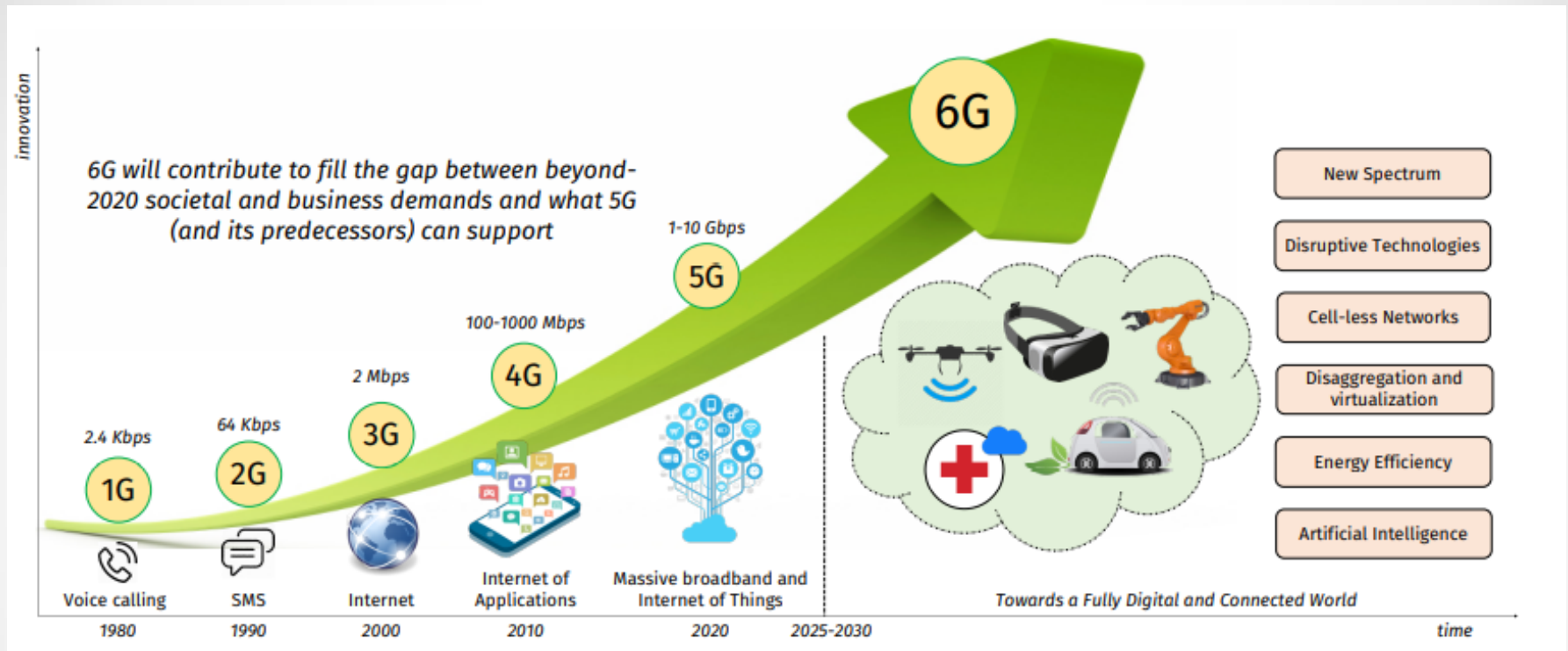


Fig. 1. Evolution of cellular networks, from 1G to 6G [1]

Introduction

5G has mainly three service scenarios

Use Cases	Requirements
Ultra-Reliable Low-Latency Communication (uRLLC)	To support low-latency transmissions of small blocks with very high reliability from a limited set of terminals
Enhanced Mobile Broadband (eMBB)	To support stable connections with very high peak data rates, as well as moderate rates for cell-edge users
Massive Machine Type Communication (mMTC)	To support massive number of Internet of Things (IoT) device

Problem & Theories

- Problem statement: 5G cope with challenges like increase in user density, seamless connectivity, traffic density, data rate, and extensive applications.

$$\text{Area throughput [bit/s/km}^2\text{]} = B \text{ [Hz]} \cdot D \text{ [cells/km}^2\text{]} \cdot SE \text{ [bit/s/Hz/cell]}$$

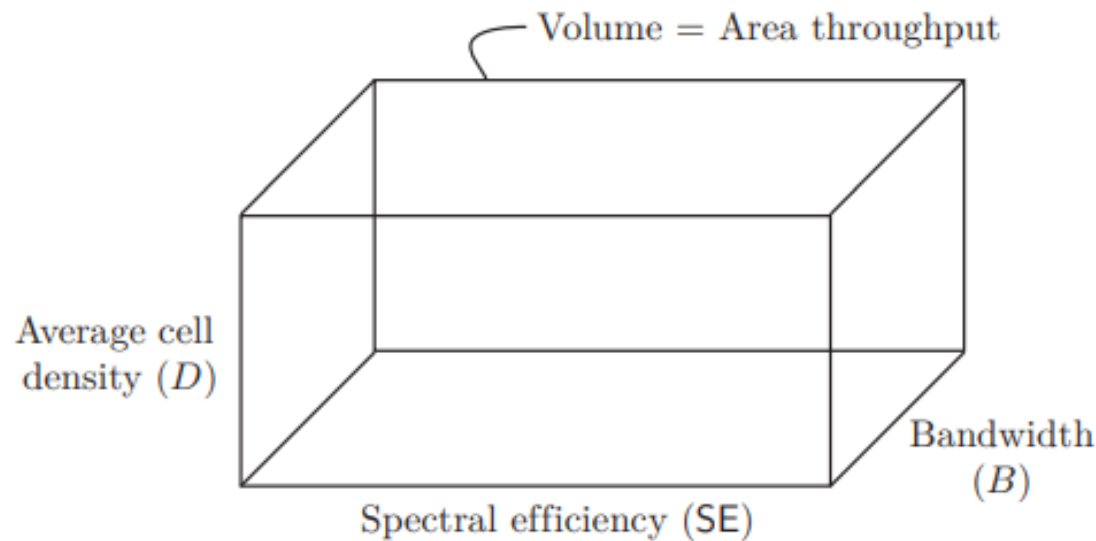


Fig.2. Three ways to improve the area throughput of cellular networks [2]

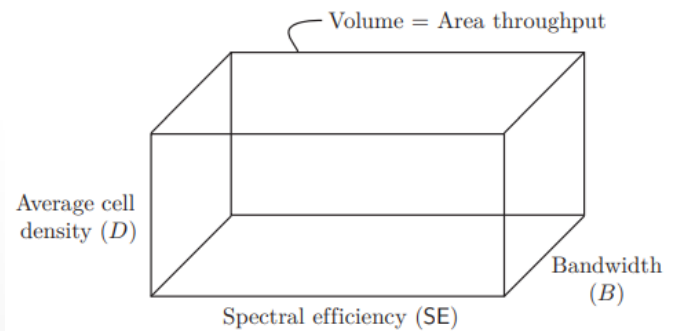
Problem & Theories

■ Possible Solutions:

1. One potential solution would be to increase the bandwidth B
2. Another potential solution would be to densify the cellular network by deploying $1000\times$ more BSs per km^2

Problems in Above Solutions: Higher cell density and larger bandwidth are approaching at saturation point where further improvements are increasingly complicated and expensive.

Ultimate Solution: Increasing the SE by virtue of new modulation and multiplexing techniques.



Methods

- To improve spectrum usage, the evolution of a new air interface and novel approaches to radio resources with multiple access management is the focus of this paper
- Orthogonal Frequency Division Multiplexing (OFDM) is a widely used multicarrier modulation air interface in 4G LTE

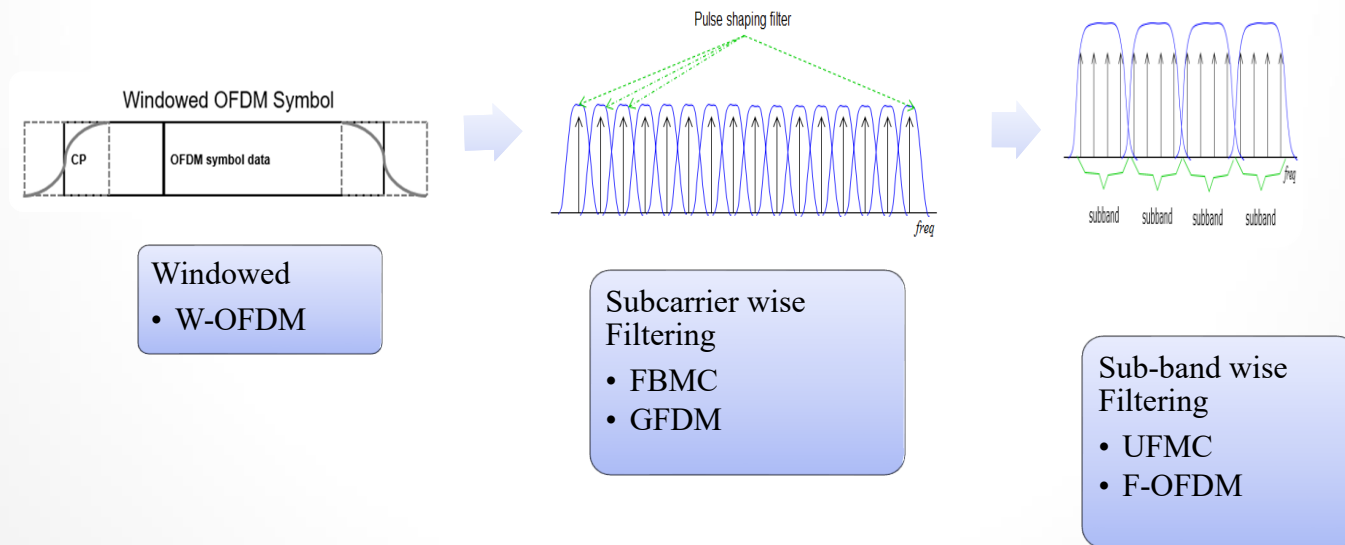


Fig. 3 Waveform Candidates for 5G

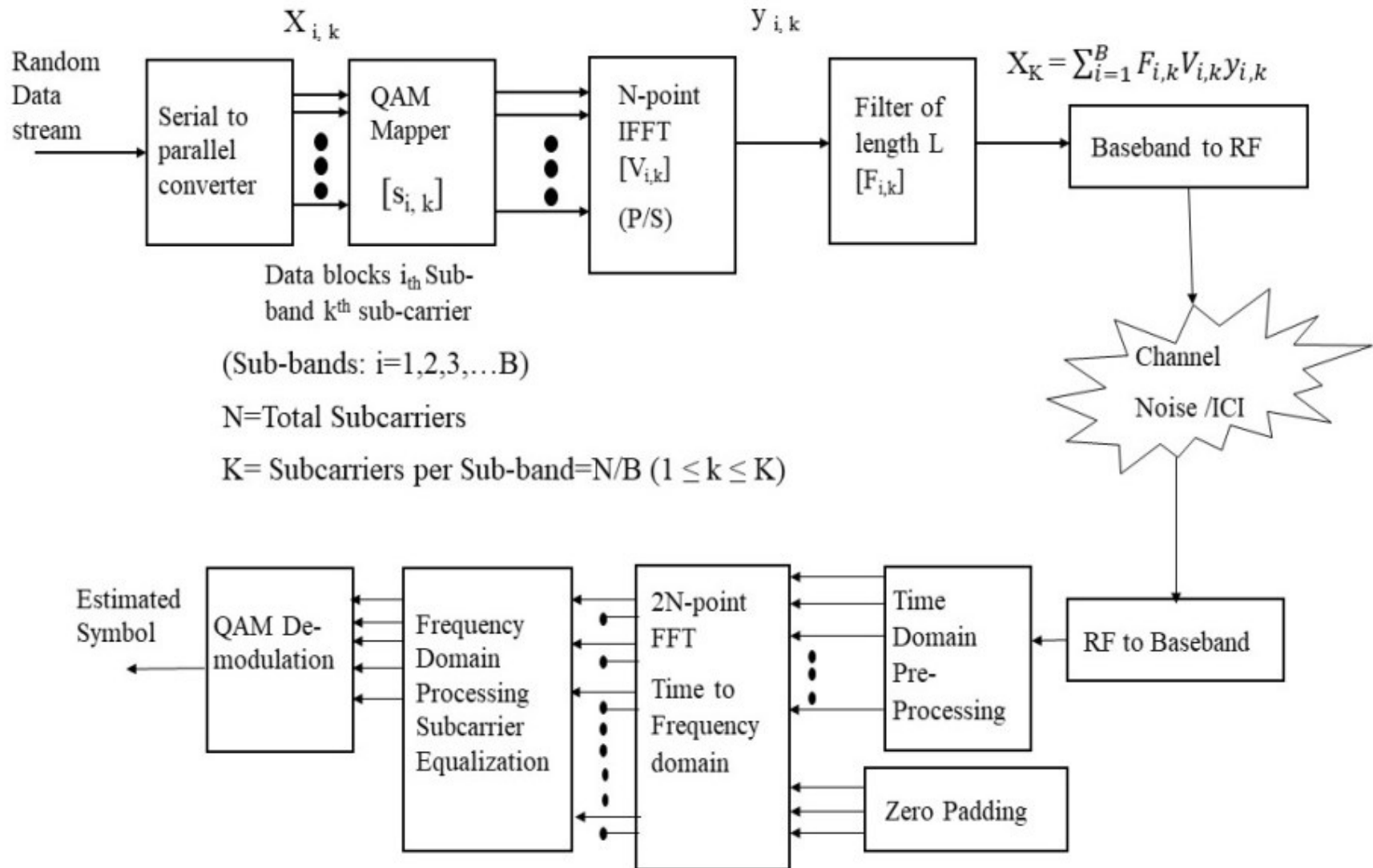
Methods

Multicarrier Technique	Details	Drawback
Filter Bank Multi-Carrier (FBMC)	Spectrum is divided into multiple orthogonal sub-bands and applies subcarrier filtering	1. Very high Computational complexity 2. Incompatibility MIMO
Generalized frequency division multiplexing (GFDM)	Modulated data symbols are transmitted two-dimensional blocks divided into sub-symbols and subcarriers which are filtered with non-orthogonal pulse shaping filters	1. Higher latency 2. Incompatibility with MIMO 3. Complex pilot design
Universal filtered multicarrier (UFMC)	Shorter filter length, and compatibility with MIMO	High PAPR



Focus of this paper

UFMC Block Diagram



Simulation Parameters

To analyze the performance of UFMC and to investigate its influence, we developed the system model in MATLAB 2021b

- The System is designed for 200 Subcarriers
- Higher order QAM modulation
- For sub-band filtering Dolph Chebyshev Filter is used

Rest other parameters are Varied to see the effect.

System Simulation and Results

1. UPMC using different sub-band schemes

Design Parameter: Allocation of the number of sub carriers per sub-band

Key Performance Indicator (KPI): Power spectral density (PSD) and Bit Error Rate (BER)

System Simulation and Results

PSD normalized spectrum of the UPMC system with varying the number of sub-carriers in each sub-band

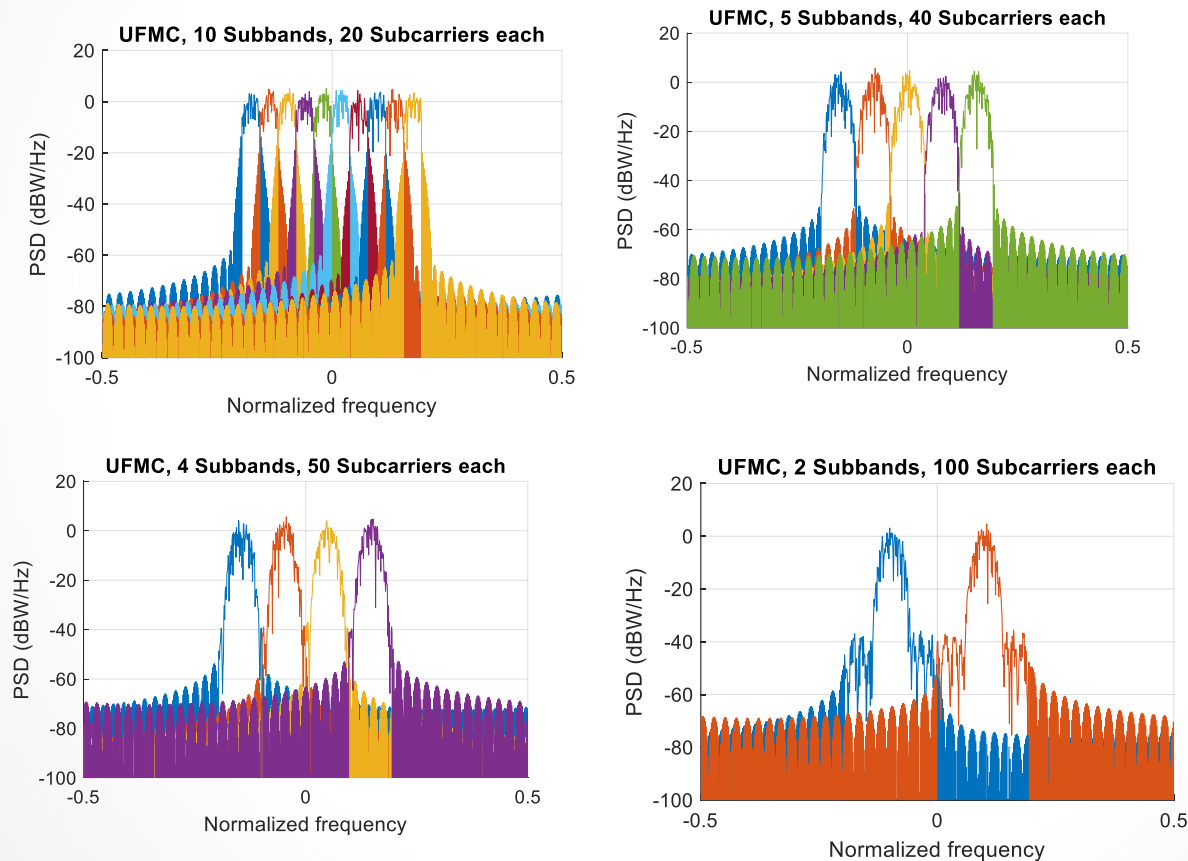


Figure 4. PSD of UPMC with different Sub-band sizes

System Simulation and Results

BER performance of UPMC with varying Sub-band size

Key Findings: The more sub-band and the smaller the sub-band size, get the better performance.

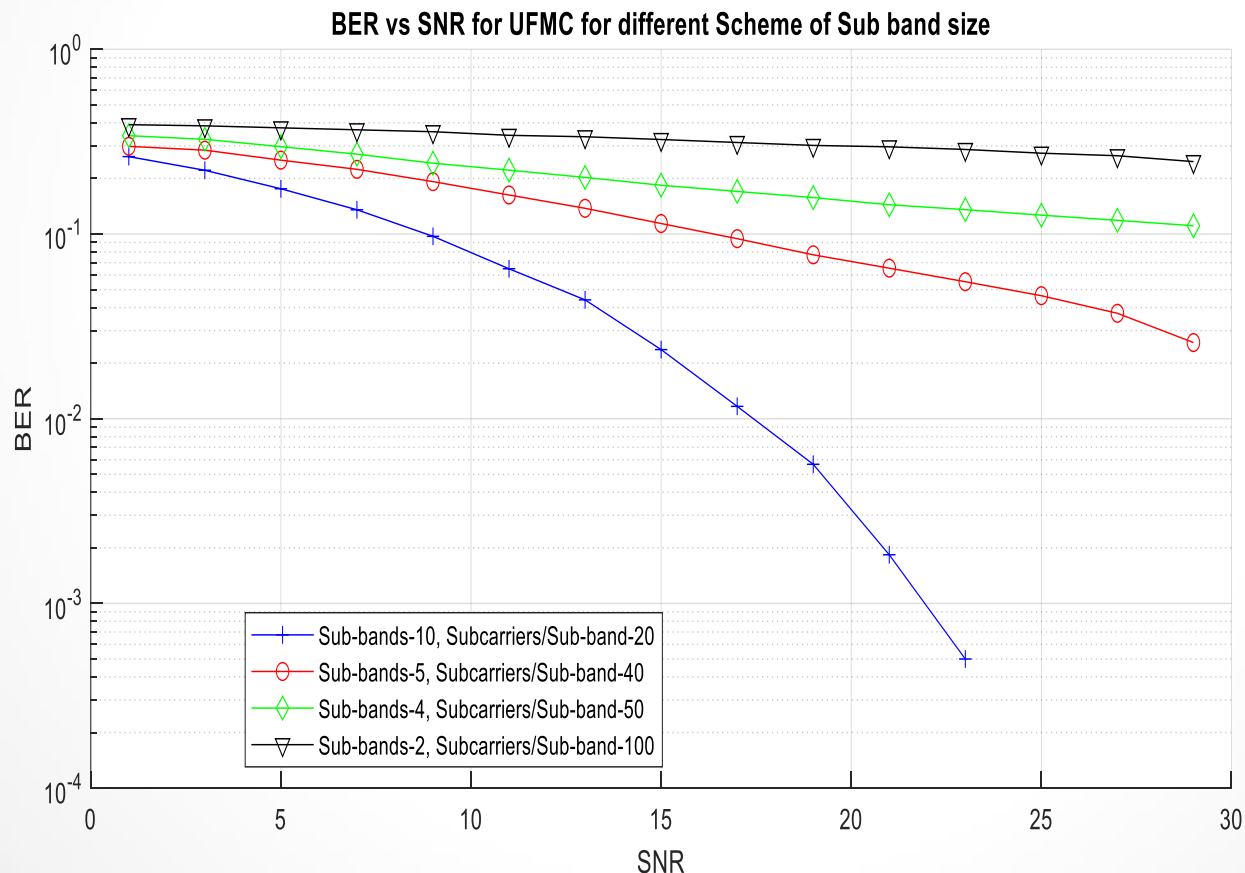


Figure 5. PSD of UPMC with different Sub-band sizes

System Simulation and Results

simulating the PSD for UPMC symbol for higher-order QAM (256 QAM) Modulation and 512-FFT, spectral efficiency increases as number of sub-carriers in each sub-band are increased

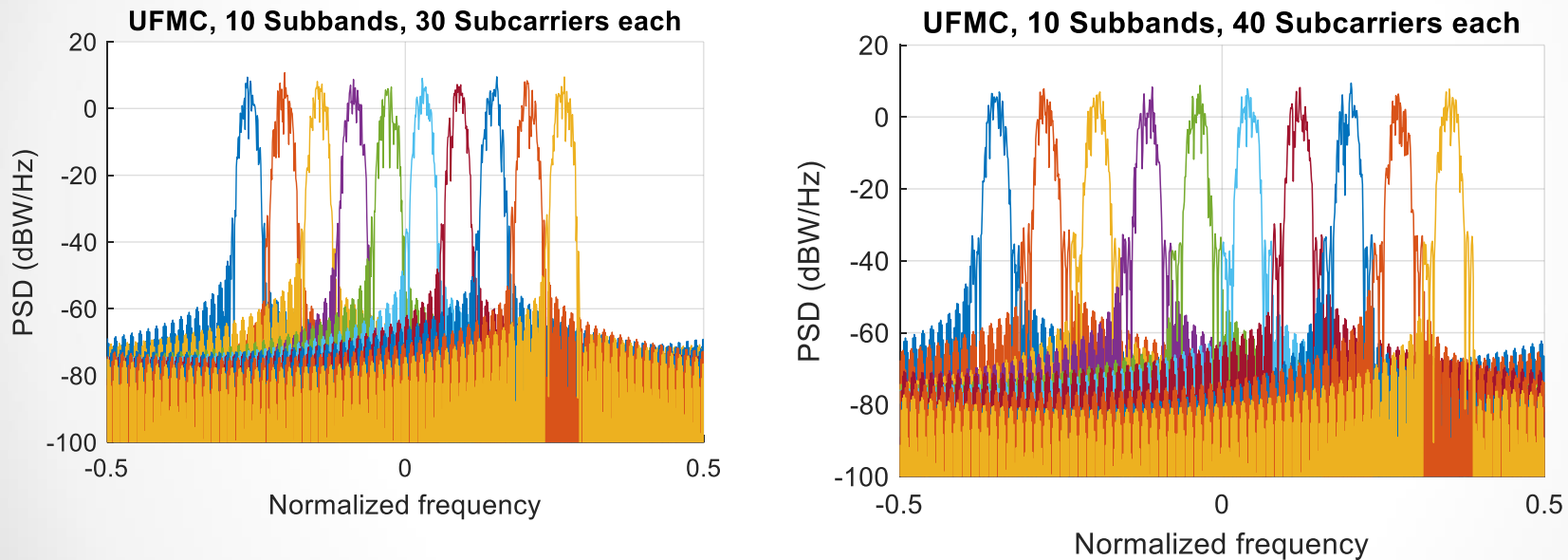


Figure 6. PSD for given Sub-band size and higher-order QAM

System Simulation and Results

2. UPMC Filter with different side lobe attenuation

Key Findings: The side lobe attenuation is increasing, the BER performance improves.

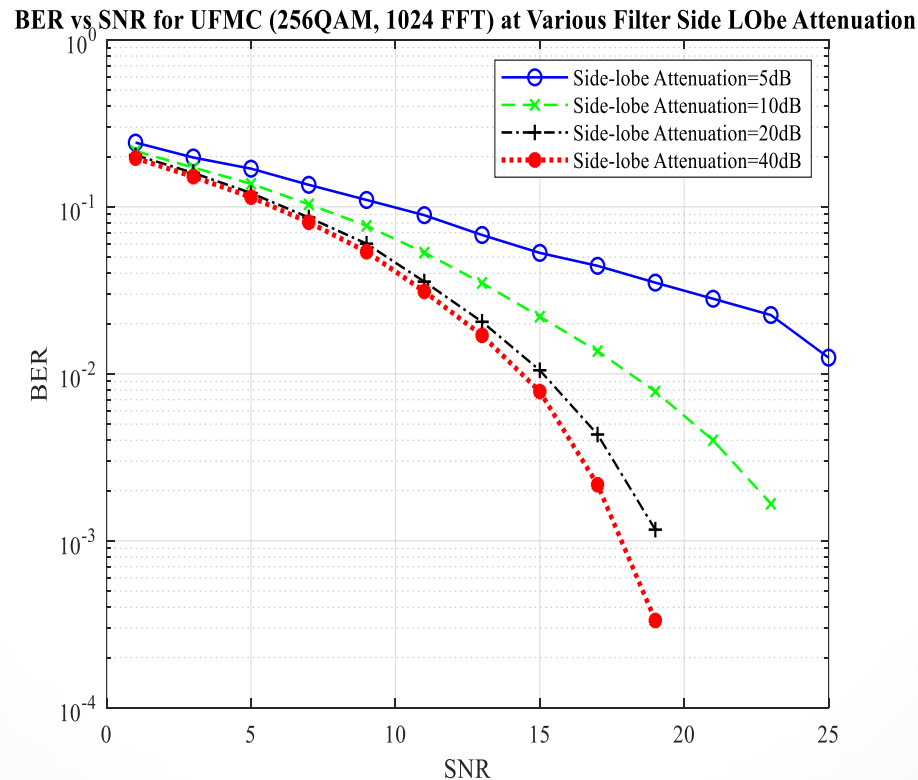


Figure 7. BER Vs SNR at different side lobe attenuation

System Simulation and Results

3. UPMC Filter with different Filter length

Key Findings: The transmission efficiency and BER performance increases as the filter length decreases

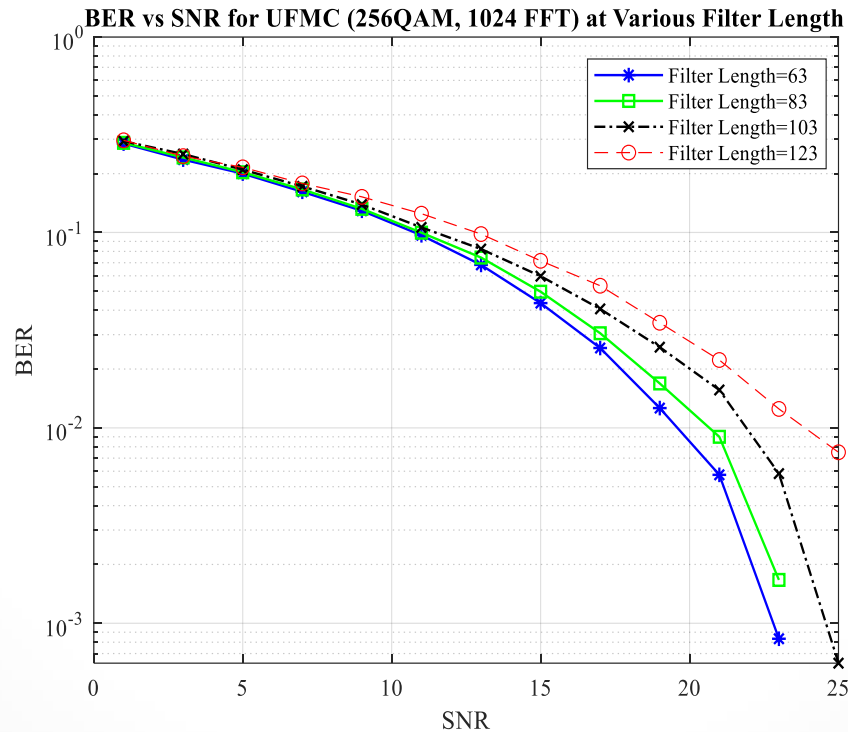


Figure 8. BER Vs SNR at different Filter length

System Simulation and Results

4. UPMC Filter with different FFT Size

Key Findings: Different FFT lengths affect the system performance in terms of CCDF, PAPR reduces with the increase in FFT size

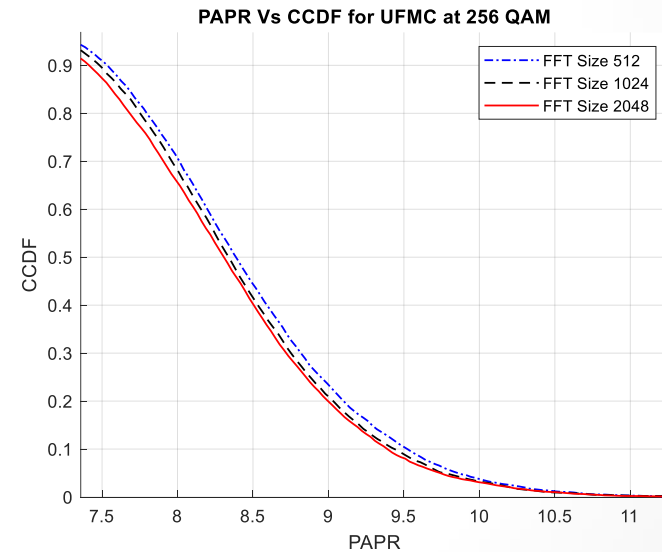
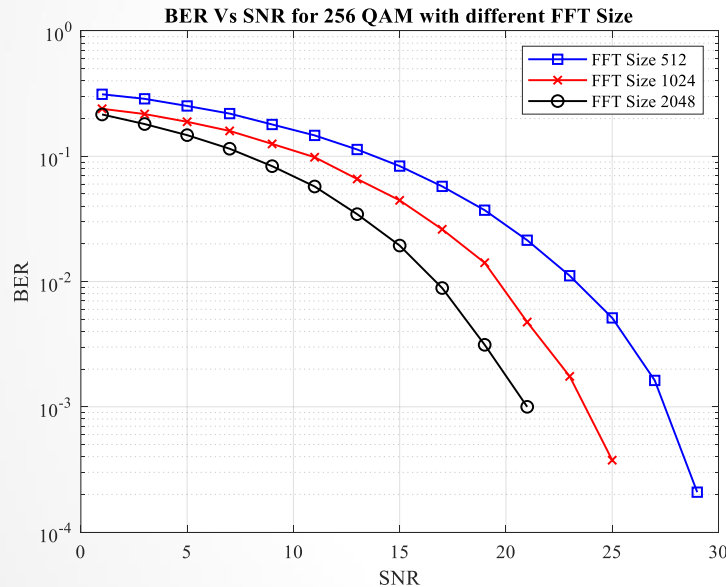
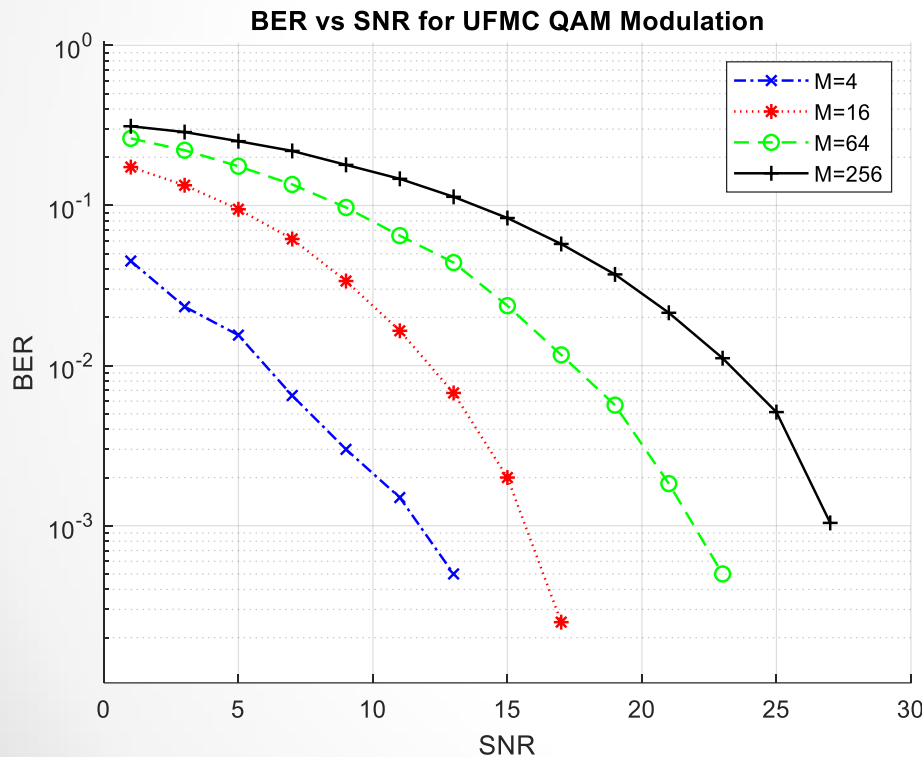


Figure 9 CCDF and BER Curve for 256QAM at different FFT Size

System Simulation and Results

5. UPMC Filter with different order of modulation

Key Findings: To achieve good frequency spectrum utilization, we need to use the higher-order QAM mapping technique. In UPMC system performance BER increases as QAM order increases.



QAM Mapping	PAPR
4 QAM	9.04 dB
16 QAM	8.2379 dB
64 QAM	8.6229 dB
256 QAM	8.0416 dB

Figure 10. BER curve for higher-order QAM

Conclusion

- This research paper concluded that when the FFT size and sub-band size increase, with side lobe attenuation and filter length decreasing, the BER performance becomes better.
- The most important result is that for the larger FFT sizes (1024 or 2048), UPMC BER performance becomes independent of FIR filter lengths.
- **Good spectral efficiency due to the absence of cyclic prefix and reduced PAPR makes UPMC better than other multicarrier techniques.**
- **Sub-band filtering helps in reducing OOB emissions**
- **Flexibility to choose sub-band size, filter length, stop-band attenuation, FFT size, and prototype window.**
- **Higher-order QAM modulation makes it best suitable with massive MIMO transmission.**

The UPMC waveform may be the best choice for the 5G and beyond wireless networks.



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THANK YOU

Questions?